SPE-227164-MS

Mapping Basement Terranes in Eastern Arabian Basin Using 3D Inversion of Magnetic Data

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This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

Abstract

Eastern Arabian Basin basement terranes are not exposed and poorly observed on seismic data due to the challenges presented by the thick Phanerozoic sedimentary cover. The identification of basement terranes has been done previously through qualitative interpretation of magnetic data. In this paper, a 3D inversion of magnetic data is used to estimate the lateral changes in magnetic susceptibilities on an interpreted depth to basement surface. The estimated magnetic susceptibilities are classified to represent the basement terranes. The top basement surface in Eastern Arabian Basin was mapped using 3D inversion of gravity data constrained by depth controls from wells and depth estimates from interpretation of magnetic data. Then another 3D inversion of magnetic data is utilized to estimate the intra-basement lateral susceptibility variations. The estimated susceptibility variations are linked directly to the magnetic mineral content of the basement layer, reflecting different terranes and being associated with significant changes in the basement composition. Phase and amplitude derivatives (tilt and theta), were applied to the estimated susceptibilities to allow for terrane boundary definition and mapping. The estimated susceptibilities were classified into 12 different unique groups based on the boundaries defined by the applied derivatives. The newly interpreted basement terranes are trending N-S, NW-SE, and NE-SW. Mean susceptibilities have been extracted for each terrane and possible rock composition have been highlighted. High susceptibility values are found in the eastern part of the basin, contradicting previous qualitative analysis. The new approach highlights possible insights for basement composition and thermal structure.

Introduction

The East Arabian Basin system is located within the Arabian Plate (Figure 1). The significant interest in studying this region arose due to the significant oil and gas resource base on one hand, and the coverage of basement structures by thick Phanerozoic sediments on the other. Located within the Arabian plate, one of the youngest continental plates on Earth (Stern and Johnson, 2010), there is a persisting challenge that rises in terms of understanding the nature of the basement, specifically basement terrane mapping. Drilled wells rarely penetrate the basement and the seismic resolution to map out the basement – sediment transition is limited (Glen et al., 2002; Direen et al., 2001).

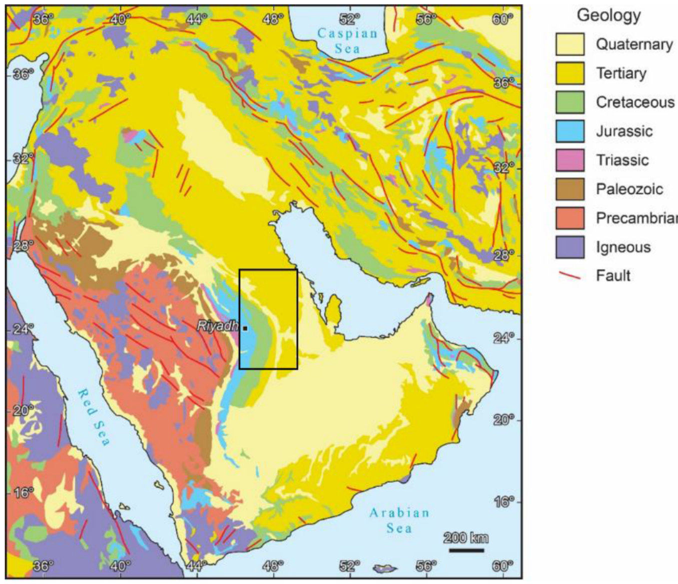

Figure 1—Map of the Arabian Plate (after Stewart, 2015) highlighting present day geology. The box indicates the location of the study area on the East Arabian Basin and delineate the boundary of the produced maps in the remaining figures.

Basement terranes are crustal fragments that were accreted together to from the tectonic plate. Each terrane is characterized by its own distinctive geological character, and the boundary between each terrane is often considered to be a suture zone or a fault (Howell, 1995; Al-Husseini, 2000; Stern & Johnson, 2010). Knowledge of basement terranes is crucial for various subsurface disciplines, such as structural geology and basin modelling, as it provides insights into the basement composition and thermal properties.

The identification of basement terranes in the Arabian Plate has been attempted previously through qualitative interpretation of potential fields data (Direen and Romine, 2007; Johnson and Stewart, 1995). Results from these studies have been correlated and integrated with petrology analysis from rock samples and cuttings acquired from available basement wells. In this paper, we attempt to map the basement terranes of Eastern Arabian Basin using a quantitative approach of inverting magnetic data into magnetic susceptibility. We use a pre-defined depth to basement surface and compute the lateral variations in magnetic susceptibilities for the depth to basement surface using 3D inversion of magnetic data. We interpret the inverted magnetic susceptibility variation as an attribute that is directly related to the geological heterogeneity of the accreted basement terranes. Terrane boundary mapping is performed by applying data enhancement and derivatives of the estimated magnetic susceptibility to allow for improved terrane boundary delineation. The values of magnetic susceptibility can be used to further understand the basement contribution to heat flow.

Geological setting

The East Arabian Basin system is located within the eastern part of the Arabian Plate, the Arabian Platform. The region is mostly characterized by an up to ~10 Km Phanerozoic sedimentary cover overlaying crystalline basement (Stern and Johnson, 2010). The western boundary consists of the exposed Precambrian Arabian Shield that took its final form with the opening of the Red Sea rifting in the Late Oligocene. The eastern plate boundary is represented by the Zagros deformation front, a thrust zone resulting from the collision of the Arabian and Eurasian plates during the Late Oligocene (Konert et al., 2001). The amalgamation of the Arabian plate began in the Neoproterozoic in the context of the Pan-African Orogeny compressional event that included island-arc and microcontinent terrane accretion (Faqira et al., 2009; Konert, et al., 2001; Stern and Johnson, 2010). This event led to the formation of the Arabian passive margin along the northern rim of Gondwana. With the end of the Precambrian, the Arabian plate witnessed the development of a series of NW-SE horsts and grabens within the Hormuz salt basin in eastern Arabia (Salem et al., 2020).

During the Precambrian to late Devonian, the Arabian plate translated from the equator to the southern hemisphere and was in an intracratonic setting by the Paleo-Tethys Ocean. The Najd Fault system (NW-SE shear fracture system) development can be traced to this period alongside numerous Precambrian rift basins (Konert et al., 2001; Salem et al., 2020). By Late Devonian - Middle Carboniferous, tectonism related to the Hercynian orogeny effected the Arabian Plate (citiation required here! E.g. Lamotte et al.). Regional uplift and extensive erosion are recorded within the Paleozoic strata. Additionally, the event is linked to the reactivation of Precambrian basement structures. In the Late Permian, a rapid translation of the plate to the equator took place. This period witnessed the closing of the Paleo-Tethys ocean and the opening of Neo-Tethys ocean as a result of breakage of the Cimmerian continental blocks (Stern and Johnson, 2010; Salem et al., 2020; Konert et al., 2001; Ziegler, 2001) The opening of the Neo-Tethys caused the Arabian plate to be part of multiple rifting stages. With the end of the Paleozoic, the Arabian plate transitioned into a passive margin setting allowing for a stable carbonate platform to be created throughout the Mesozoic. Tectonism resumed by Cretaceous with the closing of the Neo-Tethys Ocean where structural reactivation and regional unconformity (Pre-Aruma) is recorded.

Sedimentation history is well recorded within the Arabian Platform. The Arabian Platform witnesses subsidence that allows for the accommodation of the Phanerozoic sediments due to sediment loading and global sea-level variations. The sedimentation process in the Arabian Platform can be traced to mid Neoproterozoic, with Ediacaran and Early Cambrian salt basins define the base of the Paleozoic (Sharland et al., 2001; Faqira et al., 2009). Deposition of variations of siliclastics and minor carbonates is recorded from the Cambrian to Devonian before the occurrence of the Hercynian orogeny in the Carboniferous (Konert et al., 2001; Faqira et al., 2009). Post-Hercynian clastic continued to deposit until the Permian (Faqira et al., 2009). In the Mesozoic the plate witnessed the deposition of marine silliclastics, evaporates and carbonates over the subsiding platform, contributing to the development of rich petroleum system (Ziegler, 2001; Stewart, 2018).

The evolution of the Arabian Plate basement terranes is often confined to the Neoproterozoic – Cambrian time period and associated with tectonic instability from Gondwana supercontinent assembly (Stoeser & Camp, 2006; Johnson, 2006; Hargrove et al., 2006; Collins & Pisarevsky, 2005). Three continental masses are recognized by (Collins and Pisarvesky, 2005) to have collided and responsible for the basement geometry. The Arabian plate is part of the Archaean Azania Block which first collided with the Congo/ Tanzania/Bangweulu Block, initiating the formation of the East African Orogeny. The amalgamated blocks then continued to collide with India-Australia/ Mawson. Following a period of continental convergence, terrane suturing, and a series of rifting, compression and strike-slip faulting (Johnson and Woldehaimanot, 2003), the Gondwana assembly was completed with the Arabian plate being located on the eastern margin of Gondwana (Al-Husseini, 2000; Sharland et al., 2001).

Basement terranes are exposed over the Arabian Shield and are characterized as juvenile Neoproterozoic basement rocks (Nehlig et al., 2002; Stern and Johnson, 2010). Although basement terranes have been identified over the Arabian Shield and mapped, the Arabian Platform section (where the Eastern Arabian Basin is located) is not. Different opinions have been presented on the nature of the basement fabric within the Arabian Platform as either similar rocks composing the Arabian Shield extending below the Phanerozoic cover or that different and foreign rocks exist instead (Stern and Johnson, 2010). (Johnson and Stewart, 1995) observed from magnetic data a flat magnetic signature east of the Arabian shield suggesting that this signature could possibly reflect a different crustal block than that of the Arabian Shield.

Available published radiometric data reviewed by (Dixon and Golombek, 1988) show that older Neoproterozoic crust can be located on the eastern margin of the Arabian Shield. Geochronological studies conducted on the basement exposures from Oman are often used as a way to constrain the type of basement within the Arabian Platform. The data suggest that an older Neoproterozoic crust, different than the Arabian Shield, is underlying the sedimentary cover of eastern Arabia (Stern and Johnson, 2010). Early sedimentation found in Eastern Arabia also provides support for the existence of a different lithosphere signature than the Arabian shield. Magmatic activity and tectonism remained active throughout the Neoproterozoic in the Arabian Shield area. The difference in stability is possibly a direct indication of the different basement fabrics between the two regions (Sharland et al., 2001; Stern and Johnson, 2010).

Magnetic Data

The total magnetic intensity data (TMI) (Figure 2) utilized in our study was acquired through a High-Resolution Aero-Magnetic Survey (HRAM), flown in 2002 at a nominal flight altitude of 120 m over topography, along 500 m spaced traverse flight lines, oriented NE-SE. Common corrections such as diurnal, IGRF removal, leveling and reduction to the pole (RTP) were applied.

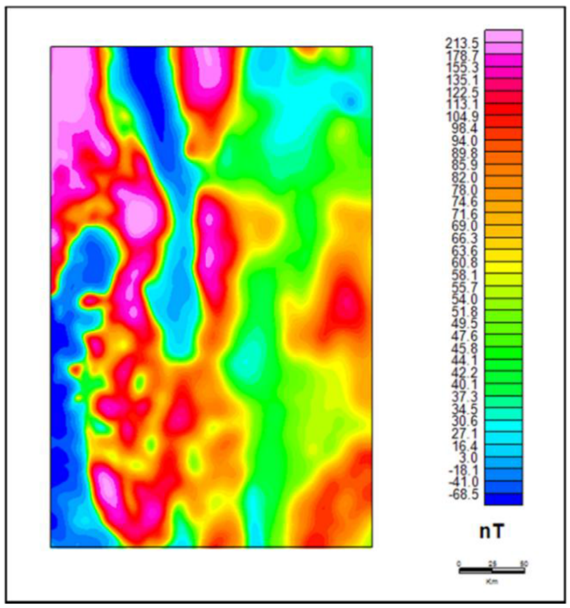

Figure 2—The magnetic anomaly map reduced to the pole of the East Arabian Basin.

The East Arabian Basin is characterized by long - wavelength anomalies towards the east with some shorter – wavelength anomalies concentrated to the west (Figure 2). The wavelength of the magnetic anomalies is a clear reflection of the depth to top-basement. Generally, long wavelength anomalies are commonly associated with deeper structural bodies, whereas shorter wavelength anomalies are an indication of shallower features (Talwani and Kessinger, 2003). Thick sedimentary cover overlaying the basement is regarded to have low impact on the magnetization due to the lack of magnetic minerals.

Magnetic data is widely used to perform qualitative interpretation to define possible basement structures and terranes. Direen and Romine (2007) performed a regional interpretation for the Arabian Plate structures and basement terranes using integration of multiple data sets including magnetic data. The correlation between each terrane and possible basement rock affinity has been concluded through the usage of basement-penetrative wells. The interpretation from (Direen and Romine, 2007) has been compared with interpreted seismic data and fault mapping conducted by (Karg et al., 2024), in which major faults and lineaments are in general agreement with the previously interpreted terranes within seismically resolvable areas.

Johnson and Stewart in their 1995 paper had previously used magnetic data over the Arabian Plate to draw a relation between the magnetic response and lineaments found with possible basement structures. The interpretation was distinguished into three categories based on the location of the magnetic anomalies observed: central, western, and eastern sectors. The central sector is defined mainly by the Central Arabian Anomaly (CAA), a large north-trending anomaly that represent the continuation of the unexposed parts of Ar Rayan composite terrane. Two other regional magnetic anomalies are identified within the central sector with unknown causative sources but could be parts of Ar Rayan terrane southern segments pre-faulting or related to intrusions. The western sector magnetic anomalies were correlated with the exposed rocks of the Arabian Shield and were mapped as terranes given the unique magnetic responses. The eastern sector represents the unexposed basement in the subsurface of the Arabian Platform. The magnetic data observed over the eastern sector is sparser in comparison to the central and western sectors. Mainly the eastern sector is characterized by a flat magnetic signature with broad-wavelength anomalies of varying intensities reflecting the deep burial of magnetic sources. No particular trend has been observed over the eastern sector except for the most western segment having north-south trending anomalies. Magnetic anomalies in the eastern sectors are believed to have been caused by crystalline rocks present in horst blocks, or intrusions.

Depth to Basement

To create a surface of depth to basement, we propose the following integrated workflow utilizing magnetic, gravity, and well data (Figure 3). Initial estimation of the depth to basement is done through multiple depth estimation steps including conventional depth determination techniques such as Euler, Werner, and spectral analysis methods (Thompson, 1982; Werner, 1953; Bhattacharyya and Leu, 1977). Seismic interpretation and well data can be used to constrain the estimation of the depth to basement from magnetic data. To further refine our depth to basement surface, 3D inversion to the gravity data was conducted and calibrated with available constraints provided from wells and magnetic data.

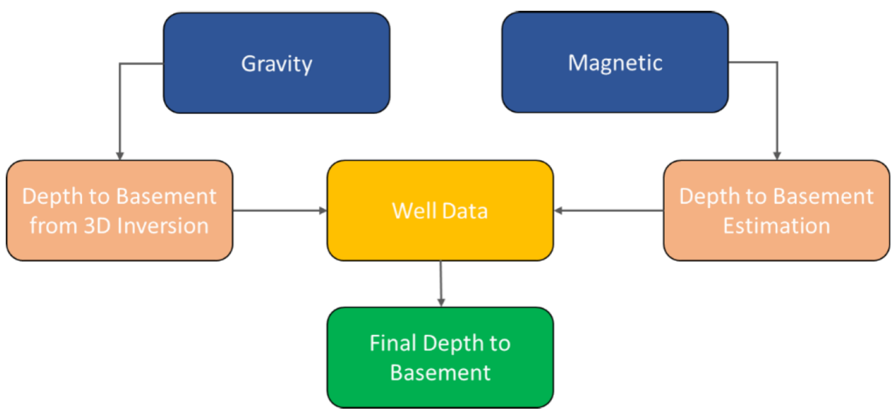

Figure 3—Depth to Basement Interpretation Workflow.

Methodology

Our methodology to map the basement terranes is based on 3D inversion of magnetic data. In our approach, we consider the magnetic field over a sedimentary basin is largely derived from the underlying magnetic basement and the information about the magnetic susceptibility is linear function of the magnetic data. Thus, in our analysis, we just need one layer with known depth to top and bottom (Figure 4). Here we describe how the depth to basement is created as well as the work flow of the 3D inversion.

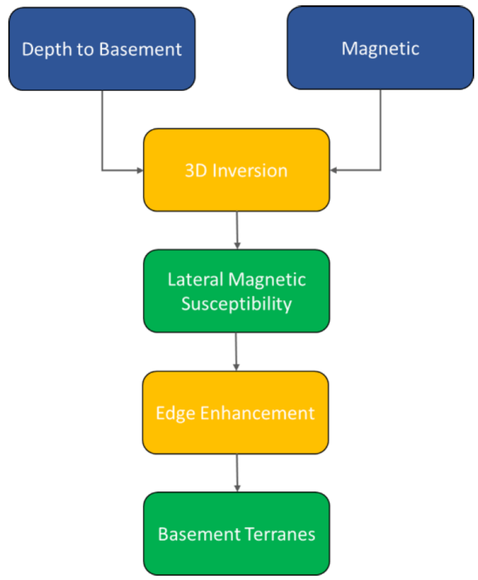

Figure 4—Basement Terrane Mapping Workflow.

Forward Modelling

The forward modelling process follows (Parker 1973) method for calculating the potential field anomalies rapidly using Fast Fourier Transform (FFT) for a non-uniform layer. The concept behind using FFT is to decrease computing time for complicated models.

(Parker 1973) first starts writing the potential fields as an expanded Taylor series around a magnetized layer. For a bounded magnetized layer, J(r), where the upper and lower boundaries are defined as z = 0 and z = h(r) respectively, the Fourier transform of the anomalous field on the plane $z = z_0$ is given by:

$$B(k) = 2\pi \mathbf{j} \cdot \mathbf{K}(\mathbf{t}) \cdot (\mathbf{K}) e^{-|k| z_0} \sum_{n=0}^{x} \frac{|\mathbf{k}|^{n-2}}{n!} F\left[J(r) h^n(r)\right], \tag{1}$$

where F[ ] represents the Fourier transformation, **j** and **t** are the magnetization and geomagnetic field directions, respectively. Since equation (1) is written in the frequency domain to allow for the rapid computation, $\mathbf{K} = (ik_x, ik_y, |\mathbf{k}|)$ is equivalent to the wave-vector **k** and it Cartesian coordinate system of $(k_x, k_y, 0)$.

A linear problem will include a flat non-varying basement surface, where h(r)=0. In this case, we can use the frequency-domain deconvolution methods to solve for magnetized layer J(r) from an observed field B(r). However, in cases of varying topography, where h(r) ≠ 0, the problem is non-linear. Thus, equation (1) will be solved through iterative methods.

Equation (1) can be linearized with a pre-defined magnetization model, $[J]_0(r)$, allowing for the resulting partial derivative matrix to have an equal size to that of the data points. To avoid running into computational issues in regards to the allowed size for the data sets, the first term in the series of equation (1) can be considered as:

$$B(k) = 2\pi \mathbf{j} \cdot \mathbf{k}(\mathbf{t}) \cdot \frac{(\mathbf{K})}{|\mathbf{k}|^2 e^{-|\mathbf{k}| z_0} F[J(r)]}, \tag{2}$$

Solving for J(r) yields:

$$J(r) = F^{-1}\left\{ \frac{|\mathbf{k}|^2 e^{-|k| z_0}}{g(k)} F[B(r)] \right\}, \tag{3}$$

Equation (3) represents deconvolution methods of magnetization mapping. F^{-1} is the inverse of Fourier transform, g(k) is the directional factors expressed in equation (2). The magnetization and field intensity share a linear relation. Source depth, shape, magnetization and geomagnetic field directions are all filtered within the frequency domain. For non-linear cases of varying basement depth, an iterative Newton-type algorithm

$$J_{n+1}(r) = \ln(r) + , \quad F^{-1}\left(\left[\frac{|k|^{-2} e^{-|\mathbf{k}|z_0}}{g(k)}\right] F\{[B_{obs}(r) - f[J_n(r)]\}\right), \tag{4}$$

is used with equation (3). J_n (r) is the estimate of J(r) at the nth iteration, B_{obs} (r) is the observed field, and $f[J_n (r)]$ represent the usage of equation (1) in the calculation of the field from the magnetization J_n (r) of the basement surface h(r). All iterative cycles are executed within the frequency domain, with the exception of the magnetization term, $J_n(r)$, is required to be computed in the space domain to ensure that the number of Fourier transforms calculations are minimized. In equation (1), the number of terms in the sum relies on the maximum value from the product of [J(r)h(r)] (Parker, 1973). If the problem is linear with no topographic variations, reducing the execution time can be done through calculating and summing the field from the magnetization correction rather than the field from the total magnetization. Thus

$$f[J_{n+1}(r)] = f[J_n(r)] + f[\Delta J_n(r)], \tag{5}$$

where $f[J_n(r)]$ is the total magnetization and $f[\Delta J_n (r)]$ is the magnetization correction. The starting estimate, $J_0(r)$, for the magnetization needed in equation (4) can be used as $J_0(r) = 0$.

The concept of inversion of potential fields data to estimate the physical properties, magnetic susceptibility or density variation, is imbedded and utilized in the GM-SYS program. (Parker and Huestis, 1974) is explicitly used to estimate the lateral magnetic susceptibility from inverting the surface magnetic data, while (Oldenburg, 1974) is used to define the method of inverting gravity data to estimate the density variations.

If we use equation (4) directly and invert, a non-unique solution would be retrieved. Therefore, according to (Parker and Huestis, 1974), equation (1) can be re-written as

$$y = f[J(r)], \tag{6}$$

where y is the observed magnetic field. An annihilator, topography-dependent magnetization, exist where

$$f[J_a(r)] = 0, \tag{7}$$

Therefore equation (6) can be finally rewritten as

$$y = f[J(r)] + cf[J_a(r)], \tag{8}$$

Where $f[J(r)]+cf[J_a(r)]$ represents the solution to equation (6), where c is a constant. The annihilator magnetization would represent a base-level shift in the magnetization. The final solution can then be written as the computed inversion of equation (4) and the addition of the multiple of the annihilator.

Detection of terrane boundaries

The edge enhancement processes depend mainly on the usage of calculated magnetic data derivatives. The calculated derivatives are used as a way to highlight the edges of basement blocks that make up the magnetic terranes through their distinguished rock property, in this case magnetic susceptibility. Two main types of derivatives are often recognized based on the behavior of the shape and amplitude above the edge location (Salem and Ali, 2015). The first type is where an inflection point, the change from negative to positive, is observed over the edge location associated with the change in basement block. An example of the first type is the tilt derivative, a technique published by Salem et al (2007), which is defined by the following equation:

$$\mathrm{TDR} = \tan^{-1}\left[\frac{\mathrm{VDR}}{\mathrm{THDR}}\right], \tag{9}$$

where VDR represent the first vertical derivative in the z direction and THDR is the total horizontal derivative defined as the root of the addition of the squared horizontal derivatives in the x and y directions.

The second type, is where the derivative would highlight the maximum absolute value at the edge location associated with the change in magnetization. The theta derivative is an example of these types of derivatives and is defined as

$$\mathrm{THETA} = \cos[\mathrm{TDR}], \tag{10}$$

In our paper, we have used both derivative types as a way to enhance our terrane delineation process.

Application

Bishop Model

Our workflow was first tested on the Bishop model. The Bishop model is recognized by SEG as an open-share data-set that allows for different geophysical methods to be tested and serves to aid in the validation of said methods (Salem et al., 2014; Fairhead et al., 2004). The data-set include the RTP magnetic map, the depth to basement, and the magnetic susceptibility map representative of the area. The model was created from a digital elevation model (DEM) for a 10.5 × 10.5 km area of the Volcanic Tablelands close to Bishop, California, that has been scaled up to produce a 3D test model with dimensions 315 × 315 km on a 500 m magnetic basement surface grid. A downward shift was applied to the topography where the depth of the highest point is a below datum by a few hundred meters making the deepest point located at a depth just above 10-km. An assumption regarding layers on top of the basement surface is made to have no magnetic values.

The approach started with calculating the observed magnetic field through forward modelling using (Parker, 1973) method that is implemented in GM-SYS 3D modelling program. In this step, our main inputs were the known lateral magnetic susceptibility grid (Figure 5c) and the pre-defined depth to basement grid for the Bishop model (Figure 5a). To build our forward model, the top layer was assigned to be the depth to basement whereas the basal layer of the model was assumed to be 30 meters, coinciding with average depth to Moho value for the Bishop Model. For the number of iterations for the Taylor Series used for the Fast Fourier Transform, the default number of five was used. The susceptibility was given by the available Bishop lateral susceptibility grid. The calculation is then carried out and produces a calculated magnetic map (Figure 5b) that reflects the magnetization signature of the susceptibility and depth to basement grids that were inputted.

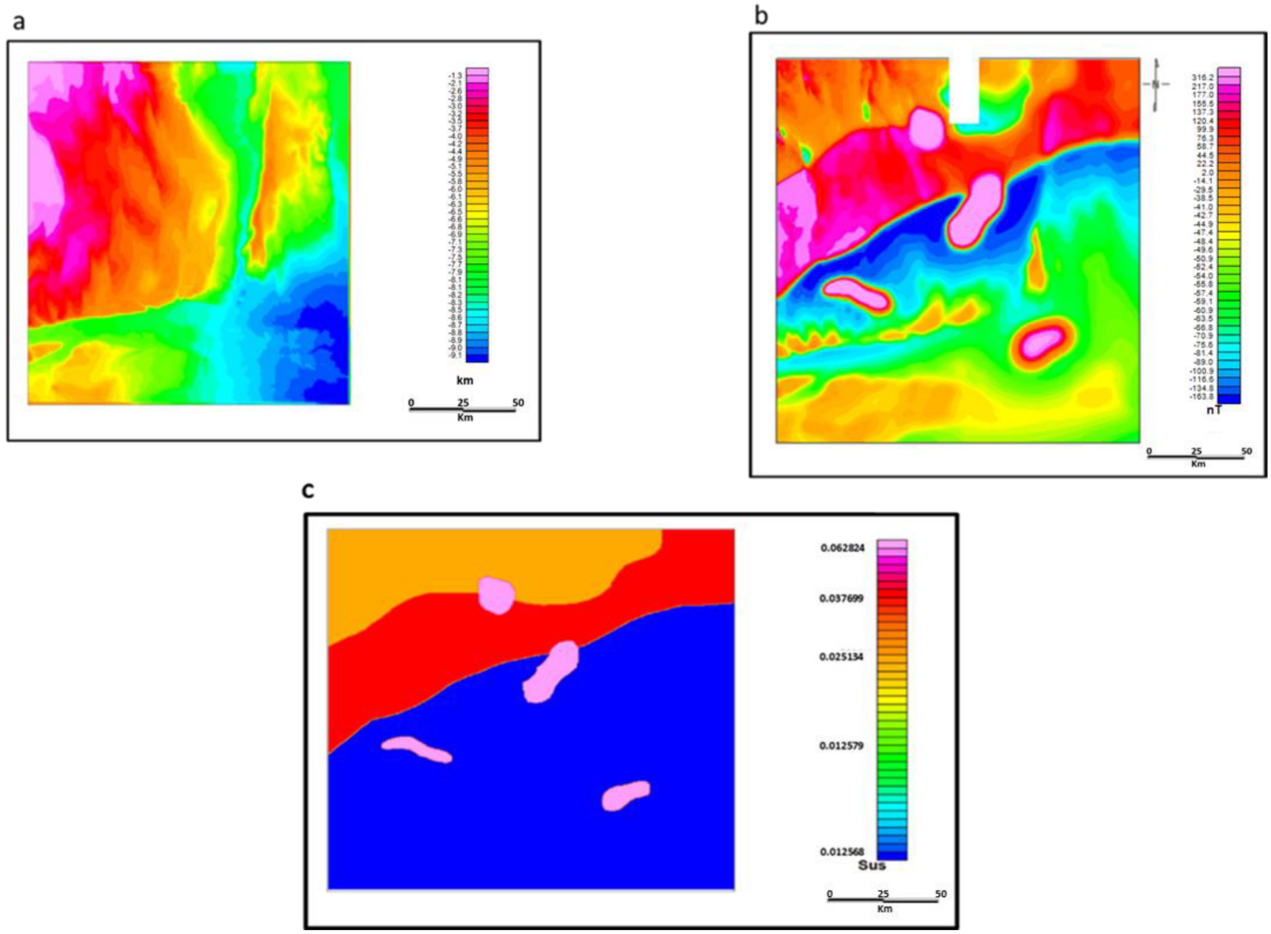

Figure 5—Bishop Model available data. (a) is the depth to basement of the Bishop Model. (b) is the observed magnetic field data of the Bishop Model. (c) is the lateral susceptibility variations grid of the Bishop Model

The resulting calculated magnetic grid is then re-inputted in the inversion as the observed magnetic grid alongside the depth to basement grid to estimate the lateral susceptibility grid. The GM-SYS 3D program utilizes two methods which can be used simultaneously to produce the susceptibility inversion, Parker – Huestis method following the (Parker and Huestis, 1974) which is excellent for long-wavelength anomalies and the Apparent Susceptibility Sharpening, to enhance short wavelength features. The recommended number of iterations for the former method is 1 – 2 iterations to ensure stability during the inversion, while its best to keep iterations for the latter less than 5 as ringing can occur with higher values. The chosen convergence limit was chosen as 10 nT, small convergence limits may be of hinderance to the model as convergence would not be reached and higher values may result in a grid with no geological sense. Performing the inversion yielded the estimated lateral susceptibilities variations for the Bishop model as shown in (Figure 6).

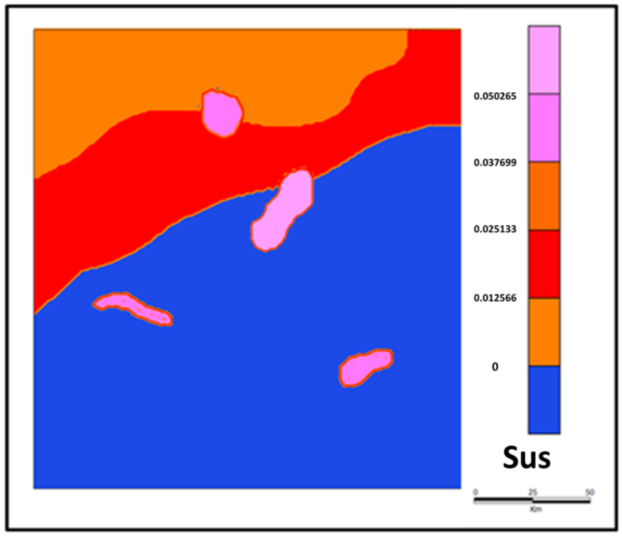

Figure 6—The estimated lateral susceptibility variations grid computed through the inversion process of the Bishop Model.

Eastern Arabia

We applied our method to the Eastern Arabia Basin in the Arabian Platform section of the Arabian Plate. Qualitative interpretation was conducted to the area by (Direen and, Romine 2007) over the area where they have also utilized their interpretation of magnetic data and segmenting the signatures into possible regional basement terranes. Over our area of interest, (Direen and Romine, 2007) show the possibility of having 8 terranes, trending N-S and NW-SE. Our approach considers the magnetic data as well, however, we focus on isolating the basement terranes through the estimated physical property.

In the case of applying the method to our area of interest, we first solved for the depth to basement surface. An initial interpretation of magnetic data was completed over the area through the application of a combination of methods including Euler, Werner, and spectral analysis. We estimated the depth to the source of each magnetic anomaly using the mentioned methods and the most coherent depth is then recorded as the solution. The results of the initial depth to basement interpretation is then constrained by available depth points from basement-penetrative wells. 3D inversion of gravity data was then applied to overcome any underestimation issues for deeper anomalies from the magnetic data interpretation. A two-layer model is built, where the top layer is the sedimentary cover and the bottom layer is the basement, separated by the constant density contrast of -0.4 g/cm^3. Then the inversion was run iteratively to achieve the optimum depth to basement grid as shown in (Figure 7).

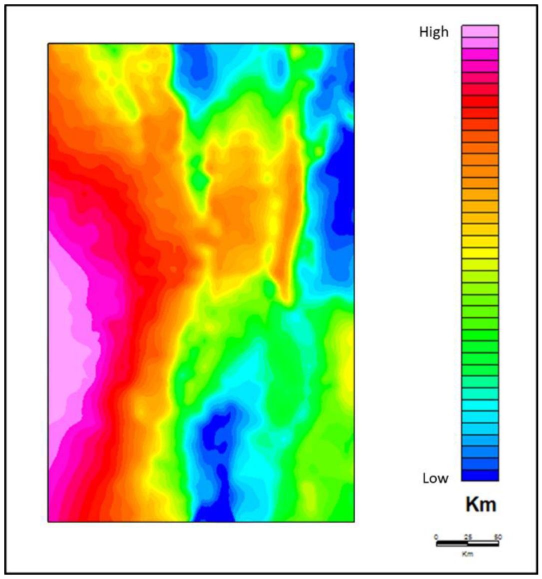

Figure 7—The depth to basement computed for the East Arabian Basin study.

To estimate the lateral susceptibility grid for the area of interest, the forward model processes serves no significant purpose as the observed magnetic field is known. Unlike the Bishop model, no known susceptibility grid is available as that would be computed through the inversion process. The model was built with the top was assigned as the previously defined depth to basement surface, whereas the model bottom was given a constant value of 40 meters in reflection of the known Moho depth in the area of interest (Mechie et al., 2013). For the susceptibility grid, a constant value of 0.01 cgi was assumed for the sake of running the forward model. Similar parameters that has been established and used during the application on the Bishop model are also used where both the Parker – Huestis and Apparent Susceptibility sharpening are used. The resulting layer (Figure 8) is the estimated lateral susceptibility variations observed over the basement of the East Arabia Basin.

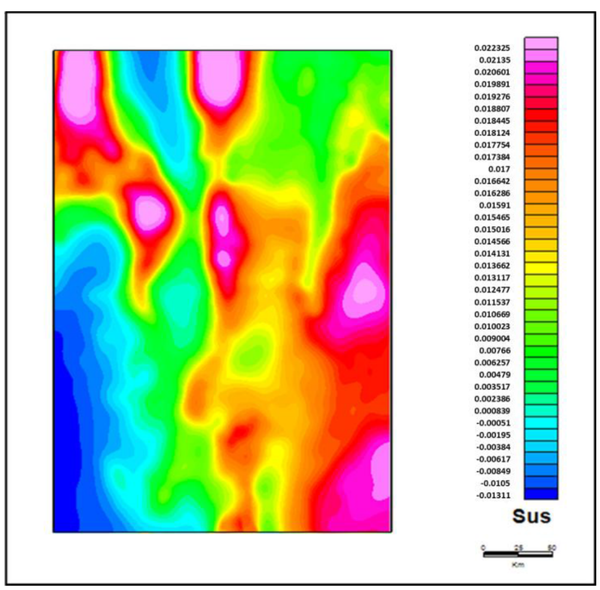

Figure 8—The estimated lateral susceptibility variations grid of the East Arabian Basin study from the inversion process.

To make sense of the computed susceptibility grid, a series of edge enhancement derivatives have been used. The objective of using the enhancing derivatives is that the lateral susceptibility variations correspond to the variations within the basement composition, hence, highlighting the edges will yield for the isolation of susceptibilities associated with basement terranes. Edge enhancement of magnetic data is often used in lineament analysis studies for the identification of basement structural variations (Salem et al., 2020). Since basement terranes are defined to be geological segments separated by sutures or faults, applying these derivatives will allow for the distinguishing of possible terrane boundaries. We applied both of the Tilt and Theta derivatives (Figure 9a and 9b) to our susceptibility grid to extract the edges for mapping. The Tilt derivative edges are mapped through locating the zero contour, whereas the mapping of the peaks will yield the Theta edges.

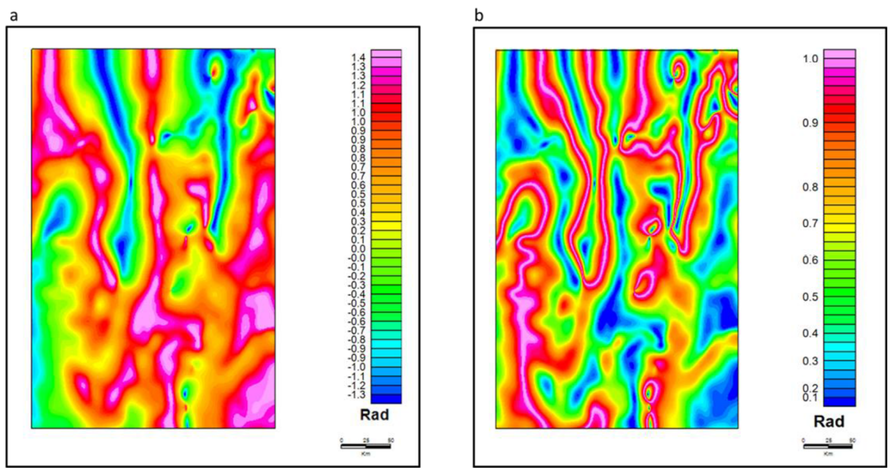

Figure 9—Edge enhancing derivatives applied to the estimated lateral susceptibility variations grid. (a) is the tilt derivative. (b) is the theta derivative.

Combining these two derivatives would outline different structural elements and aid in providing a more coherent interpretation (Figure 10). The final mapping resulted in 12 terranes that were separated based on the edge enhancement derivatives.

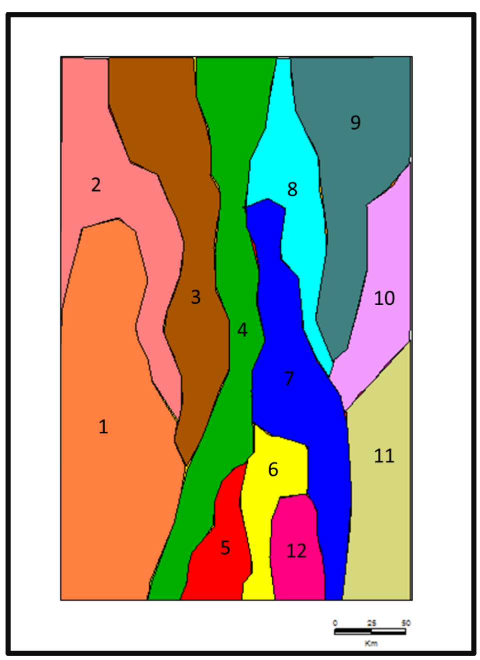

Figure 10—Final Amended Terranes based on the quantitative interpretation of the lateral susceptibility grid from the utilization of edge enhancement derivatives.

Discussion and Conclusions

We have attempted to study the basement terranes in the East Arabian Basin. Our approach depended on formulating a workflow that investigates the physical property of magnetic susceptibility rather than inferring descriptively from magnetic data. The workflow included defining the depth to basement surface of the area of interest, calculating susceptibility variations through magnetic inversion, and applying edge enhancement techniques to isolate basement susceptibility variations into unique terranes. The methodology was applied first to the known Bishop Model to estimate the necessary model parameters to be used on our field data. Forward modelling was first conducted on the Bishop Model to solve for the observed magnetic field, later used in the inversion process. The purpose of computing the forward model is to verify that the output magnetic field would correspond to the input susceptibility grid, establishing a one to one relationship and maintaining the integrity of the inverse process.

Observed magnetic data (Figure 5b) retrieved from the forward process on the Bishop Model is highly similar to the actual provided data. Additionally, results from the applied inversion to solve for the lateral magnetic susceptibility variations are excellent when compared to the actual Bishop Model data. The estimated susceptibility grids for the Bishop Model and field data show a nice correlation to that of the input observed magnetic field grids (Figure 5c). The correlation corresponds to the direct proportional relationship magnetization has with susceptibility, given that the geomagnetic field strength remains constant. The contrast seen in the resulting outputs is a consequence that arises from using Fast Fourier Transform on a variable layer. The usage of FFT in magnetization mapping to isolate geological boundaries by forming a direct relation with rock properties is usually done under the assumption of non-varying basement layer. As seen with (Parker, 1973) and (Parker-Huestis, 1974), while the purpose is to solve for magnetization in varying topography, the equation is still linearized to fit the concept of a non-varying surface. In order to apply a more accurate approach for a varying layer, the usage of space-domain methods is recommended for magnetization mapping as outlined by (Misener et al., 1984). The usage of FFT, however, is still considered to be optimal due to the decreasing in computation time needed to process large data-sets.

We have mapped the depth to basement surface (Figure 7) across the basin through constructing a 3D inversion model of gravity data. The model was constrained by depth controls from basement – penetrative wells and depth estimates from magnetic data interpretation. Initial observations of the magnetic data (Figure 2) show that we expect deeper structures within the eastern part of the basin due to the long – wavelength anomalies seen. The computed depth to basement surface demonstrates a significant increase in depth of the Precambrian basement rocks from west to east. This increase in depth towards the east is an evident feature of the Arabian Plate's asymmetry agrees with the published information (Al-Husseini, 2000; Mechie et al., 2013; Stern and Johnson, 2010; Faqira et al., 2009). It is possible that this asymmetry could be attributed to lithospheric buoyancy variations as a result of crustal heterogeneity between the Arabian Shield and Arabian Platform (Stern and Johnson, 2010; Johnson and Stewart, 1995).

The estimated lateral susceptibility variations grid (Figure 8) highlights a wide range of magnetic susceptibilities found within the basin's basement. The wide distribution and different clusters observed are a representation of the heterogeneous nature of the basement composition as a result of its accretion and evolution. While magnetic susceptibility is a direct measure to the amount of magnetic mineral found within the geology (Li et al., 2021), another factor that needs to be considered is how magnetic susceptibility varies with temperature. Magnetic susceptibility generally decreases with increased temperature and all magnetism is lost beyond the Curie Isotherm depth where temperature reaches around 540 ̊C (Ross et al., 2006). Based on this relationship, we can expect lower susceptibility measurements in deeper regions as temperature is directly proportional with depth and rocks are closer to reaching Curie Isotherm depth. However, the depth to Curie's Isotherm is variant and is dependent of the overall thermal structure of the crust (Salem et al., 2024).

Basement terrane interpretation was conducted through the help of edge enhancement techniques provided by magnetic data derivatives applied to the lateral susceptibility grid. Derivatives over magnetic data has often yielded excellent interpretation of basement lineaments related to possible structures and features given that the basement fabric is highly magnetized in contrast to the sediments overlaid (Salem et al., 2020). Initial interpretation was conducted on the lateral susceptibility grid and was found to be highly similar and in agreement with the previous qualitative interpretation. From only using the susceptibility grid, 12 terranes (Figure 9) were identified that trend N-S and NW-SE. With the help of edge enhancement, these 12 terranes were further interpreted and amended. Final amended terranes are referenced in Figure 10 and the mean susceptibility and standard deviation of each terrane was extracted (Table 1).

Table 1—Extracted Magnetic Susceptibility values for the interpreted basement terranes of Eastern Arabian Basin. Give unit to the values to be 10-6 SI

Terrane Number	Mean Susceptibility	Standard Deviation
1	0,004674	0,007065
2	0.018564	0.003367
3	0.011837	0.003258
4	0.017405	0,003475
5	0.01399	0.002218
6	0.017031	0.000774
7	0.016838	0.001111
8	0.015389	0.001117
9	0.014816	0.001526
10	0.020038	0 001537
11	0.0197	0.001177
12	0.015972	0.000854

Previous qualitative interpretation conducted on the area utilized core samples from basement-penetrative wells to optimize the magnetic interpretation of the basement terranes (Direen and Romine, 2007). It was deduced that as we head from west to east, the basement composition becomes more felsic. However, the number of wells analyzed for the eastern terranes are few and sparse to support generalizing that the whole eastern region of the study area to be of felsic material. Additionally, the previous data concluded that the wells may have only reached the infracambrian layer above the basement unlike what has been seen from wells within the central and western portion of the study area. Our susceptibility estimates support the possibility of basement rocks of mafic affinity existing given the cluster of high values of susceptibility found in the east. Nonetheless, our estimates within the central and western region are in general agreement with the previous qualitative interpretation core analysis where ultramafic to intermediate rocks seem to exist in combination of metasediments.

According to (Vitorello & Pollack, 1980), heat flow in continental crustal types is mainly attributed to radiogenic heat flow that is determined through the crust rock composition and age. Additional heat comes from tectonothermal activities and from the deep lower crust and mantle sources. Heat production is highest in felsic rocks such as granite and rhyolite because of the high content of radiogenic material. Whereas basalts and mafic rocks have low radiogenic heat contribution (Rybach, 1976). Based on this established relationship, magnetic susceptibility can be used indirectly to optimize heat flow values. Since magnetic susceptibility decreases with increased temperatures and heat, high radiogenic heat flow values would correspond to lower magnetic susceptible basement rocks, possibly of felsic composition. By looking at the magnetic susceptibility values, we should expect low contribution of heat from the basement.

Mapping basement terranes using inversion of magnetic data seems to be promising with respect to qualitative interpretation approaches. This approach is more effective as it depends solely on the rock physical property rather than inferring descriptively from the magnetic anomaly data. The approach is highly beneficial as it allows for understanding the basement geometry further aiding in building structural frameworks. Furthermore, utilizing magnetic susceptibilities can help in inferring for the petrological characteristics of the basement and its implication on heat flow analysis.